UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

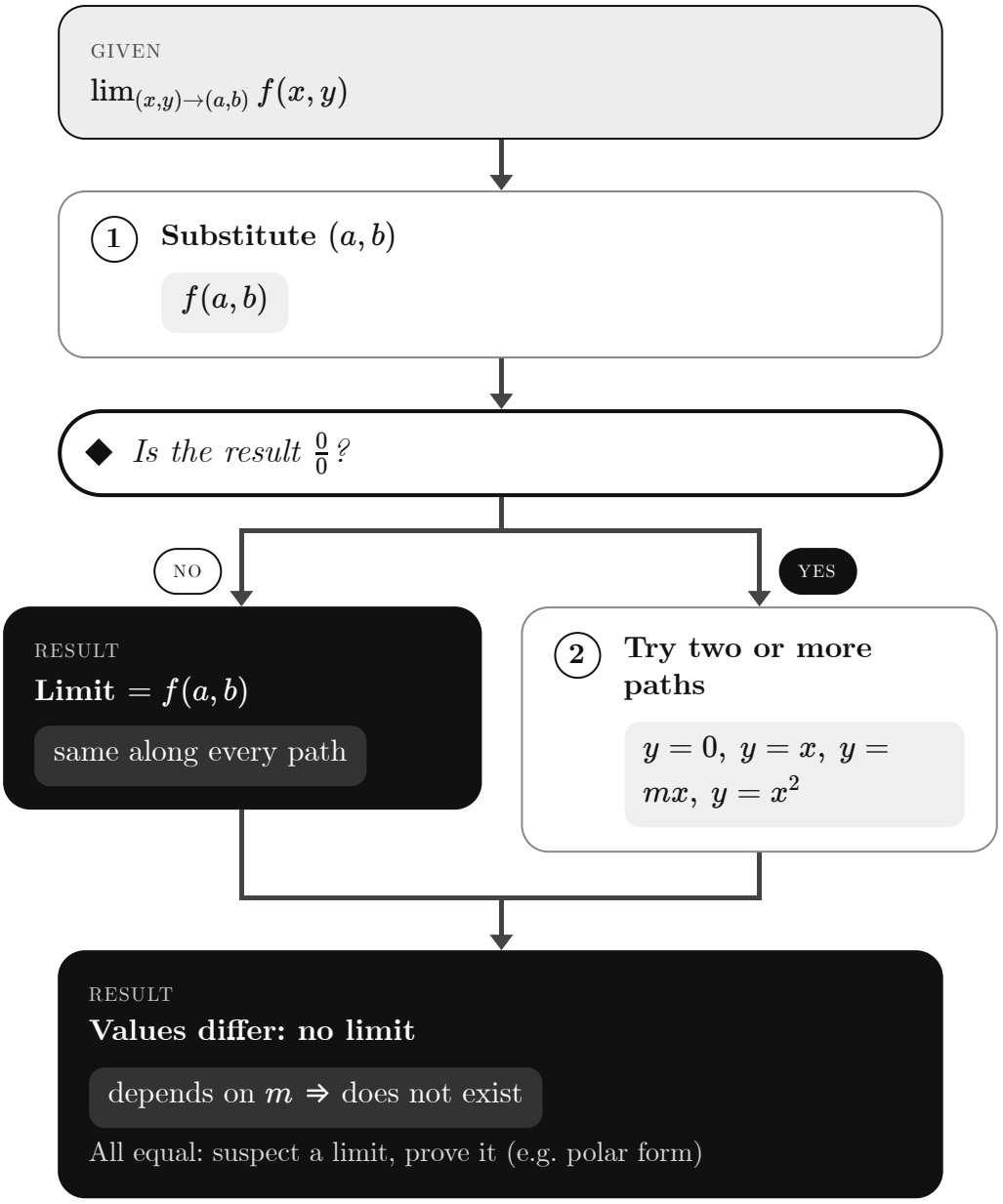
Class 001 · Introduction to partial differentiation · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a

Contents

1. Figure F2.1 · Finding (or disproving) a limit at a point
2. Figure F3.1 · Is f continuous at (a, b)?
3. Figure F4.1 · Computing a partial derivative
4. Figure F5.1 · Reading the order of a mixed partial
5. One variable and two variables

2 Limit of a function of two variables



Remark. Agreement along a few paths is only evidence; disagreement is proof.

Exam trap. Read the point carefully: slide $(2,1)$ gives -8, the lecture's $(2,-1)$ gives -6.

Figure F2.1. Finding (or disproving) a limit at a point ESA ★★

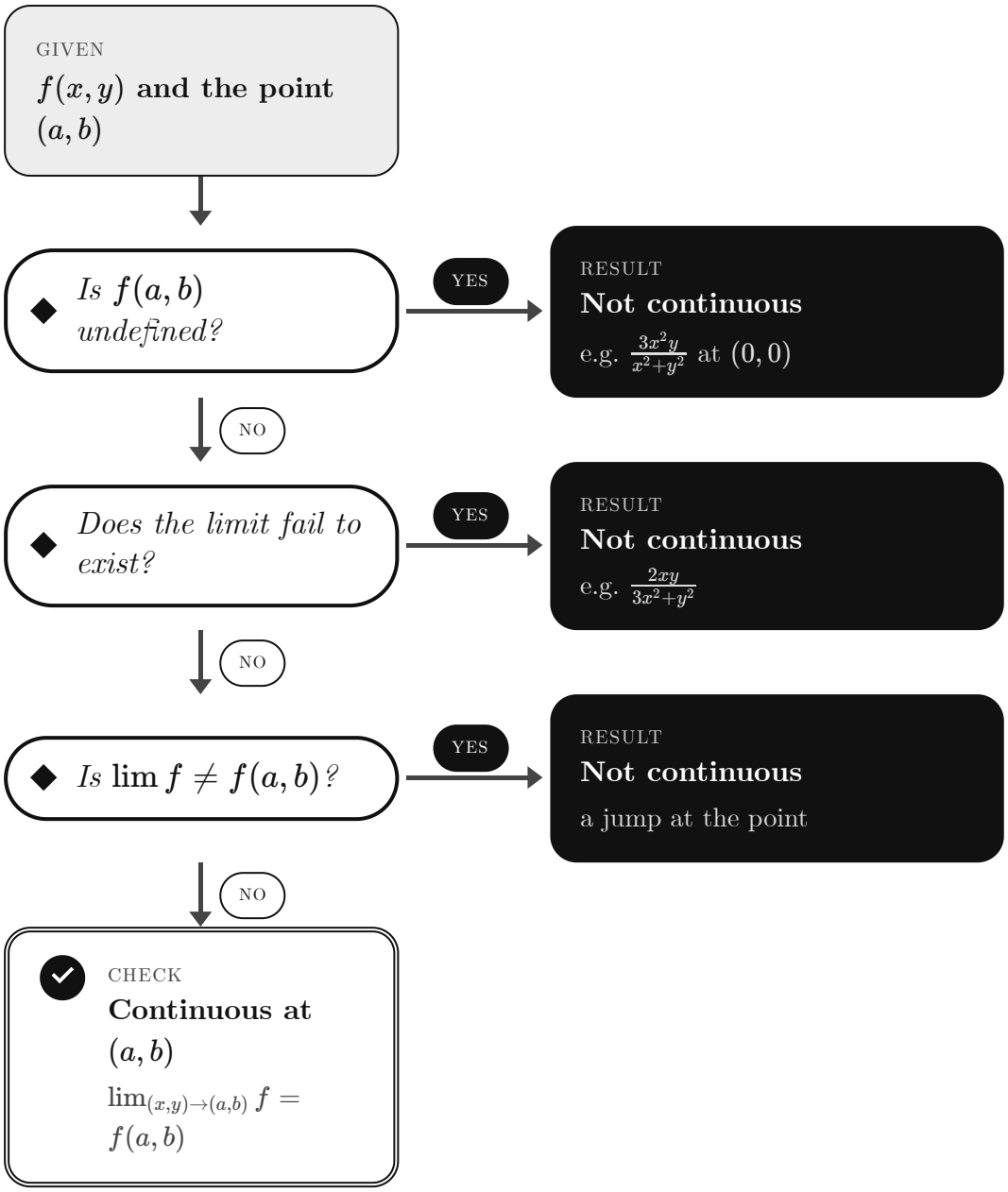
S001·9 · S001·10 · T001a 00:13:59 · T001a 00:17:42

One variable and two variables

	$y = f(x)$	$z = f(x, y)$
Domain	Real numbers	Ordered pairs (x, y)
Graph	A curve in the plane	A surface in space
Approach to the point	From the left or the right	Along infinitely many paths
Limit exists if	Left and right limits agree	Every path gives the same value
Continuous if	$\lim_{x \to c} f = f(c)$	$\lim_{(x,y) \to (a,b)} f = f(a, b)$
Derivative	$\frac{dy}{dx}$	$\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ (others constant)

S001·9 · S001·11 · T001a 00:09:54 · T001a 00:11:52

3 Continuity of a function of two variables



Remark. Only the first rung failing with the limit existing is a removable discontinuity: define $f(a,b)$ as the limit.

Figure F3.1. Is f continuous at (a, b)? ESA ★

S001·11 · T001a 00:18:46 · T001b 00:03:17

4 Partial derivatives: history, definition and notation

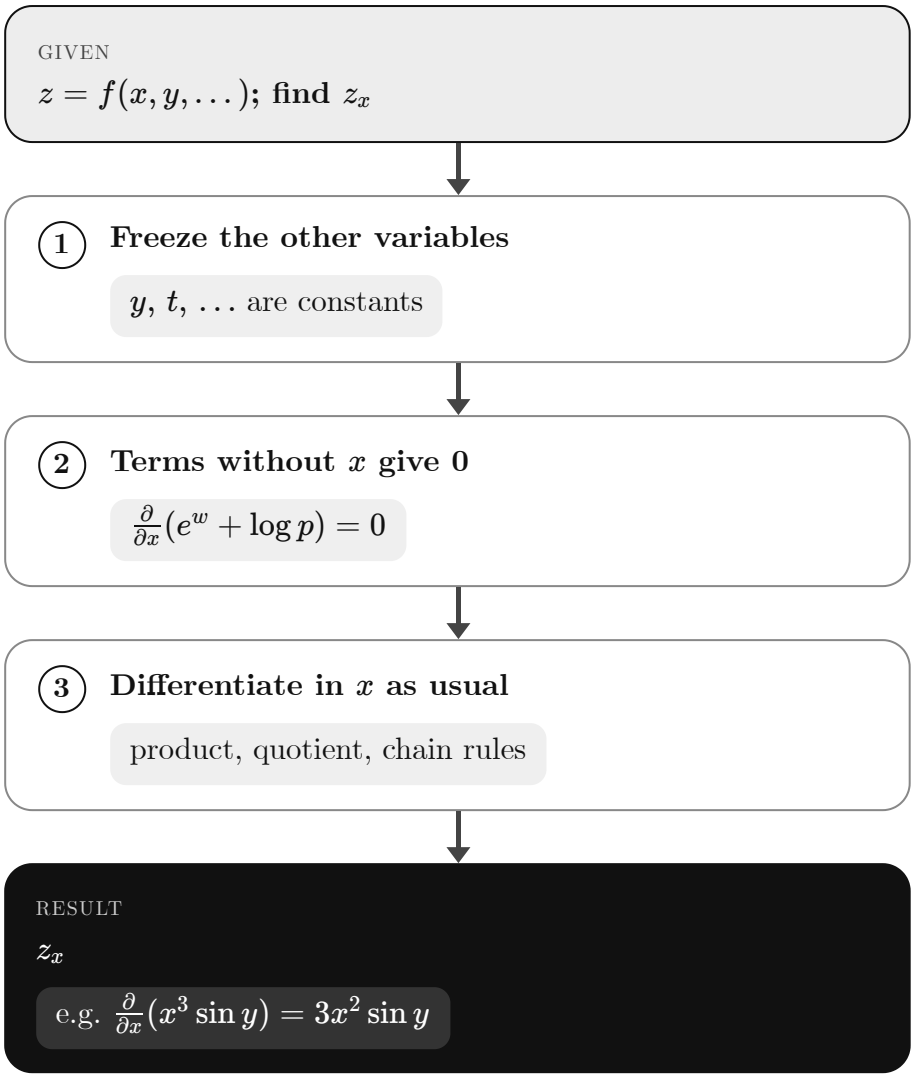
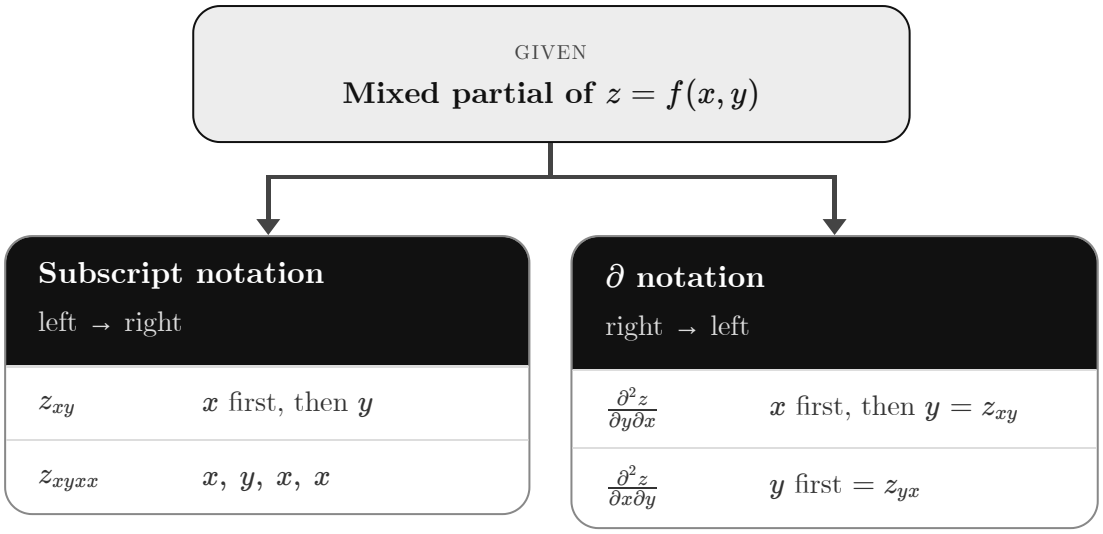


Figure F4.1. Computing a partial derivative ESA ★★

S001·15 · T001b 00:15:08 · T001b 00:13:57

5 Second-order and mixed partial derivatives



Remark. By Clairaut's theorem the order rarely changes the answer, but the notation is still examined.

Figure F5.1. Reading the order of a mixed partial ESA ★★

S001·17 · T001b 00:18:51